System Parameters

- θ : Angle at which the bike frame is inclined with respect to the vertical.
- ψ : Angle of the reaction wheel frame with respect to the bike frame.
- M : Mass of the bike frame (excluding the reaction wheel).
- M_w : Mass of the reaction wheel.
- H : Height of the bike frame center of gravity from ground level.
- H_w : Mounting height of the reaction wheel from ground level.
- I_1 : Moment of inertia of the bike frame (excluding the wheel) about the pivot point (ground).
- I_w : Moment of inertia of the reaction wheel about its spin axis.

The kinematics are formulated using Lagrangian mechanics.

Kinetic Energy

1. Rotational Energy of the Bike Frame

The rotational kinetic energy of the bike frame about the ground pivot is

$$T_{\text{frame}} = \frac{1}{2} I_1 \dot{\theta}^2.$$

2. Rotational Energy of the Reaction Wheel

The reaction wheel undergoes:

1. Rotation about its own spin axis with angular velocity

   $$\dot{\theta} + \dot{\psi},$$

   giving kinetic energy

   $$\frac{1}{2} I_w (\dot{\theta} + \dot{\psi})^2.$$

2. Rotation about the ground pivot due to the motion of the bike frame. Using the parallel-axis theorem,

   $$\frac{1}{2} M_w H_w^2 \dot{\theta}^2.$$

Therefore, the total kinetic energy of the reaction wheel is

$$T_{\text{wheel}} = \frac{1}{2} I_w (\dot{\theta} + \dot{\psi})^2 + \frac{1}{2} M_w H_w^2 \dot{\theta}^2.$$

Total Kinetic Energy

Combining both contributions,

$$T = \frac{1}{2} \left(I_1 + M_w H_w^2 \right) \dot{\theta}^2 + \frac{1}{2} I_w (\dot{\theta} + \dot{\psi})^2.$$

Potential Energy

The gravitational potential energy of the bike frame and reaction wheel is

$$V = (MH + M_w H_w)\, g \cos\theta.$$

Lagrangian

The Lagrangian is defined as

$$L = T - V.$$

Substituting the expressions for kinetic and potential energy,

$$L = \frac{1}{2}\left(I_1 + M_w H_w^2\right)\dot{\theta}^2 + \frac{1}{2} I_w(\dot{\theta} + \dot{\psi})^2 - (MH + M_w H_w)\, g \cos\theta.$$

Euler–Lagrange Equations

We have two generalized coordinates, θ and ψ, so the Euler–Lagrange equation is applied twice. The general form of the Euler–Lagrange equation is

$$\frac{d}{dt}\left(\frac{\partial L}{\partial \dot{q}}\right) - \frac{\partial L}{\partial q} = Q_q,$$

where q is the generalized coordinate and Q_q is the corresponding generalized force.
We first apply the equation to θ and then to ψ.

Equation 1: Euler–Lagrange Equation for θ

We first calculate

$$\frac{\partial L}{\partial \dot{\theta}}.$$

Treating $\dot{\theta}$ as the variable, we differentiate every term containing $\dot{\theta}$:

$$\frac{\partial L}{\partial \dot{\theta}} = \left(I_1 + M_w H_w^2\right)\dot{\theta} + I_w(\dot{\theta} + \dot{\psi}).$$

Taking the time derivative gives

$$\frac{d}{dt}\left(\frac{\partial L}{\partial \dot{\theta}}\right) = \left(I_1 + M_w H_w^2\right)\ddot{\theta} + I_w(\ddot{\theta} + \ddot{\psi}).$$

We next calculate

$$\frac{\partial L}{\partial \theta}.$$

Only the potential-energy term contains a bare θ. Therefore,

$$\frac{\partial L}{\partial \theta} = (MH + M_w H_w)\, g \sin\theta.$$

The generalized force corresponding to θ is

$$Q_\theta = 0.$$

Therefore, the Euler–Lagrange equation becomes

$$\frac{d}{dt}\left(\frac{\partial L}{\partial \dot{\theta}}\right) - \frac{\partial L}{\partial \theta} = 0.$$

Substituting the expressions obtained above,

$$\left(I_1 + M_w H_w^2\right)\ddot{\theta} + I_w(\ddot{\theta} + \ddot{\psi}) - (MH + M_w H_w)\, g \sin\theta = 0.$$

Hence,

$$\boxed{\left(I_1 + M_w H_w^2\right)\ddot{\theta} + I_w(\ddot{\theta} + \ddot{\psi}) = (MH + M_w H_w)\, g \sin\theta} \tag{1}$$

Equation 2: Euler–Lagrange Equation for ψ

We now apply the Euler–Lagrange equation to the second generalized coordinate, ψ.

First,

$$\frac{\partial L}{\partial \dot{\psi}} = I_w(\dot{\theta} + \dot{\psi}).$$

Taking the time derivative,

$$\frac{d}{dt}\left(\frac{\partial L}{\partial \dot{\psi}}\right) = I_w(\ddot{\theta} + \ddot{\psi}).$$

There is no term in the Lagrangian that contains a bare ψ. Therefore,

$$\frac{\partial L}{\partial \psi} = 0.$$

The generalized force corresponding to ψ is the motor torque,

$$Q_\psi = \tau.$$

Therefore, the Euler–Lagrange equation becomes

$$I_w(\ddot{\theta} + \ddot{\psi}) = \tau.$$

Hence,

$$\boxed{I_w(\ddot{\theta} + \ddot{\psi}) = \tau} \tag{2}$$

This equation describes the angular acceleration of the reaction wheel relative to the bike frame.

Generalized Forces and Motor Torque

The difference between Q_θ and Q_ψ is important.

The generalized force corresponding to a coordinate is determined by the virtual work associated with a small virtual displacement of that coordinate:

$$\delta W = Q_q \, \delta q.$$

The reaction-wheel motor is mounted between the bike frame and the reaction wheel. Its torque acts across the relative angular coordinate ψ.

Therefore, the motor torque contributes directly to the generalized force associated with ψ:

$$Q_\psi = \tau.$$

There is no direct external torque associated with the coordinate θ, so

$$Q_\theta = 0.$$

The reaction torque acting on the bike frame is therefore not inserted manually. It appears automatically when equations (1) and (2) are combined.

Simplifying the Equations of Motion

From equation (2),

$$I_w(\ddot{\theta} + \ddot{\psi}) = \tau.$$

Therefore,

$$I_w(\ddot{\theta} + \ddot{\psi}) = \tau.$$

Substituting this directly into equation (1),

$$\left(I_1 + M_w H_w^2\right)\ddot{\theta} + \tau = (MH + M_w H_w)\, g \sin\theta.$$

Rearranging,

$$\boxed{\left(I_1 + M_w H_w^2\right)\ddot{\theta} = (MH + M_w H_w)\, g \sin\theta - \tau} \tag{3}$$

Therefore, the angular acceleration of the bike frame is

$$\boxed{\ddot{\theta} = \frac{(MH + M_w H_w)\, g \sin\theta - \tau}{I_1 + M_w H_w^2}} \tag{4}$$

Equation (4) is the main equation describing the rotational dynamics of the bike frame.
From equation (2), we can also obtain the angular acceleration of the reaction wheel.
Starting with

$$I_w(\ddot{\theta} + \ddot{\psi}) = \tau,$$

divide both sides by I_w:

$$\ddot{\theta} + \ddot{\psi} = \frac{\tau}{I_w}.$$

Therefore,

$$\boxed{\ddot{\psi} = \frac{\tau}{I_w} - \ddot{\theta}} \tag{5}$$

Substituting equation (4) into equation (5),

$$\ddot{\psi} = \frac{\tau}{I_w} - \frac{(MH + M_w H_w)\, g \sin\theta - \tau}{I_1 + M_w H_w^2}.$$

Hence,

$$\boxed{\ddot{\psi} = \frac{\tau}{I_w} - \frac{(MH + M_w H_w)\, g \sin\theta - \tau}{I_1 + M_w H_w^2}} \tag{6}$$

Equations (4) and (6) completely describe the rotational dynamics of the bike frame and reaction wheel.

Physical Interpretation

Equation (4) can be understood in terms of the torques acting on the bike frame.
The gravitational torque is

$$\tau_g = (MH + M_w H_w)\, g \sin\theta.$$

This torque tends to increase the inclination of the bike away from the upright position.
The reaction wheel produces an opposing torque on the bike frame,

$$\tau_{\text{reaction}} = -\tau.$$

Therefore, the net torque responsible for the angular acceleration of the bike frame is

$$\tau_{\text{net}} = (MH + M_w H_w)\, g \sin\theta - \tau.$$

The effective rotational inertia of the bike frame, including the contribution of the reaction-wheel mass, is

$$I_{\text{eff}} = I_1 + M_w H_w^2.$$

Thus, equation (3) can be interpreted simply as

$$I_{\text{eff}}\ddot{\theta} = \tau_{\text{net}}.$$

The important point is that the reaction wheel does not need to push against the ground to generate a corrective torque. The motor accelerates the reaction wheel relative to the bike frame. Due to conservation of angular momentum, an equal and opposite reaction torque acts on the bike frame.

This is the fundamental principle behind reaction-wheel stabilization.

Equilibrium Condition

We can now determine the motor torque required to maintain the bike at a constant inclination angle.

For equilibrium,

$$\dot{\theta} = 0$$

and

$$\ddot{\theta} = 0.$$

Substituting $\ddot{\theta} = 0$ into equation (3),

$$0 = (MH + M_w H_w)\, g \sin\theta - \tau.$$

Therefore, the required motor torque is

$$\boxed{\tau = (MH + M_w H_w)\, g \sin\theta} \tag{7}$$

This means that the motor torque must exactly balance the gravitational torque.

For the perfectly upright position,

$$\theta = 0.$$

Since

$$\sin 0 = 0,$$

equation (7) gives

$$\boxed{\tau = 0.}$$

However, the upright position is not automatically stable.

If the bike is disturbed by a small angle, gravity produces a torque that tends to increase the disturbance rather than reduce it. Therefore, feedback control is required to continuously adjust the reaction-wheel torque and return the bike to the upright position.

Small-Angle Approximation

For control-system analysis, the bike is generally considered to operate close to the upright position. Therefore, θ is small and the small-angle approximation can be used:

$$\sin\theta \approx \theta.$$

Using this approximation in equation (3),

$$\left(I_1 + M_w H_w^2\right)\ddot{\theta} = (MH + M_w H_w)\, g\theta - \tau.$$

Rearranging,

$$\boxed{\left(I_1 + M_w H_w^2\right)\ddot{\theta} - (MH + M_w H_w)\, g\theta = -\tau} \tag{8}$$

Dividing through by the effective rotational inertia,

$$\boxed{\ddot{\theta} = \frac{(MH + M_w H_w)\, g}{I_1 + M_w H_w^2}\theta - \frac{1}{I_1 + M_w H_w^2}\tau} \tag{9}$$

This equation makes the instability of the upright position clear.

The coefficient multiplying θ is positive:

$$\frac{(MH + M_w H_w)\, g}{I_1 + M_w H_w^2} > 0.$$

Therefore, if θ becomes positive, the gravitational term produces a positive angular acceleration. Similarly, if θ becomes negative, the gravitational term produces a negative angular acceleration.

Thus, a disturbance tends to grow rather than decay.

The reaction-wheel torque τ must therefore be controlled so that it opposes this gravitational effect.

State-Space Representation

For control-system analysis, it is convenient to express the equations of motion in first-order state-space form.

We define the state vector as

$$\boxed{\mathbf{x} = \begin{bmatrix} \theta \\ \dot{\theta} \\ \psi \\ \dot{\psi} \end{bmatrix}}$$

and define the motor torque τ as the control input:

$$u = \tau.$$

The first state equation follows directly from the definition of angular velocity:

$$\boxed{\dot{x}_1 = x_2}$$

The second state equation is obtained from equation (4):

$$\boxed{\dot{x}_2 = \frac{(MH + M_w H_w)\, g \sin x_1 - u}{I_1 + M_w H_w^2}}$$

The third state equation is

$$\boxed{\dot{x}_3 = x_4.}$$

Finally, from equation (5),

$$\dot{x}_4 = \frac{u}{I_w} - \dot{x}_2.$$

Substituting the expression for $\dot{x}_2$,

$$\boxed{\dot{x}_4 = \frac{u}{I_w} - \frac{(MH + M_w H_w)\, g \sin x_1 - u}{I_1 + M_w H_w^2}}$$

Therefore, the complete nonlinear state-space model is

$$\boxed{\dot{\mathbf{x}} = \begin{bmatrix} x_2 \\ \dfrac{(MH + M_w H_w)\, g \sin x_1 - u}{I_1 + M_w H_w^2} \\ x_4 \\ \dfrac{u}{I_w} - \dfrac{(MH + M_w H_w)\, g \sin x_1 - u}{I_1 + M_w H_w^2} \end{bmatrix}} \tag{10}$$

where

$$\mathbf{x} = \begin{bmatrix} \theta & \dot{\theta} & \psi & \dot{\psi} \end{bmatrix}^T$$

and

$$u = \tau.$$

For operation close to the upright position, using

$$\sin\theta \approx \theta,$$

the state-space model becomes linear:

$$\dot{\mathbf{x}} = A\mathbf{x} + Bu.$$

The system matrix is

$$\boxed{A = \begin{bmatrix} 0 & 1 & 0 & 0 \\ \dfrac{(MH + M_w H_w)g}{I_1 + M_w H_w^2} & 0 & 0 & 0 \\ 0 & 0 & 0 & 1 \\ -\dfrac{(MH + M_w H_w)g}{I_1 + M_w H_w^2} & 0 & 0 & 0 \end{bmatrix}}$$

and the input matrix is

$$\boxed{B = \begin{bmatrix} 0 \\ -\dfrac{1}{I_1 + M_w H_w^2} \\ 0 \\ \dfrac{1}{I_w} + \dfrac{1}{I_1 + M_w H_w^2} \end{bmatrix}}.$$

This linearized model can now be used for controller design techniques such as PID, state feedback, or Linear Quadratic Regulator (LQR) control.